Fiscal Year:	FY 2010	**Task Last Updated:**	FY 08/06/2010
PI Name:	Oman, Charles M. Ph.D.		
Project Title:	Validation of Assessment Tests and Countermeasures for Detecting and Mitigating Changes in Cognitive Function During Robotics Operations		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Neurobehavioral and Psychosocial Factors Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02139-4301	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Crew Health NNJ08ZSA002N
Start Date:	07/01/2009	**End Date:**	06/30/2013
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Czeisler, Charles (Brigham and Women's Hospital) Liu, Andrew (Massachusetts Institute of Technology) Lockley, Steven (Brigham and Women's Hospital) Barger, Laura (Brigham and Women's Hospital)		
Grant/Contract No.:	NCC 9-58-NBPF02001		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The goals of this four-year NASA/university project are: (1) characterize the changes in cognitive function during robotic operations that affect performance, (2) validate proxy cognitive assessment tests such as WinSCAT or PVT as predictors of performance changes in a complex operational task, and (3) test the efficacy of fatigue countermeasures (e.g., light, caffeine, modafinil) to improve cognition during robotic operations. To date, the complex robotics operations aboard the Shuttle and ISS have been successful, but not entirely without incident. The impairment of cognitive abilities, likely due to fatigue, has been identified as a factor leading to errors in performance. In nominal operations, pauses, unnecessary movements and performance inefficiency may reflect cognitive impairment. Preventative countermeasures are necessary to reduce this risk and improve crew safety for current and future robotic operations. In this project, subjects will be trained to perform simulated robotics tasks under realistic astronaut schedules. Using a within-subjects design, our experiments will investigate how the proposed fatigue countermeasures affect both cognitive and task performance, thereby enabling their relationship to be understood. If this correlation can be established, we will provide more evidence that these assessment tools could be used as indicators of fitness-for-duty. The project will be a collaborative effort between the MIT Man Vehicle Laboratory (MVL), the Division of Sleep Medicine at Brigham and Women's Hospital (BWH) and Harvard Medical School, the NASA Johnson Space Center Mechanical and Robotics Systems Group, and the NASA Astronaut Office. MIT will develop the simulations of robotics operations and implement the cognitive assessment tests. The BWH/Harvard team will create realistic schedules and environmental conditions to test the impact of countermeasures under these conditions. We will work closely with NASA trainers and astronauts to guide the direction of the research and recreate valid operational scenarios.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	An effective, safe, well-tolerated, non-invasive countermeasure for circadian- and fatigue-related deficits in cognition is required for use during in space environments to enhance the safety of crew members. Light exposure has the potential to fulfill this role. Although monochromatic blue light exposure at night has been shown to be most effective at shifting the circadian pacemaker, and improving alertness and performance, these results have not been tested for broadband blue-enriched white light or for light exposure during the biological day. Before light therapy as fatigue countermeasure is operationalized, further research is required to fully understand the effect on complex performance, such as robotics performance. Similarly, while caffeine use is widespread, including on the ISS, uncontrolled use of caffeine may not be optimally timed or of the correct dose to alleviate sleepiness on duty and may interfere with subsequent sleep, thereby increasing fatigue the next day. The current study will investigate the effects of light and caffeine as fatigue countermeasures in a controlled environment with the long-term view to developing specialized countermeasure schedules to maximize alertness and performance of space crew in an environment where a even a small fatigue-related errors could have catastrophic consequences. These results will also have widespread application on Earth. Approximately 18,000 Air Force pilots and flight crew are exposed to conflicting circadian signals and resulting fatigue from long flights. Tens of thousands of troops are subjected to rapid deployment world-wide and are often required to maintain alertness while suffering from jet lag. Additionally, nearly 180,000 troops are required to be vigilant for more than 24 hours at a time, and thousands of Navy and Coast Guard personnel work for extended periods under circadian misalignment. Over 200,000 commercial airline pilots, flight crews and air traffic controllers are also subjected to extended shifts and circadian misalignment. More than 230,000 ER doctors, staff and medical residents work extended shifts for 30 hours or more. Approximately 1.9 million public safety personnel (police, corrections officers, firefighters, etc.) work extended shifts and are exposed to emergencies which require vigilance for long periods. In total, more than 3 million Americans would directly benefit from the findings of this research. In addition, applications may be possible in standard occupational settings to improve general day-time alertness; in educational settings such as in training sessions and conferences, colleges, and schools where enhanced alertness would assist learning and memory; as a fatigue countermeasure for sleepy car and truck drivers in order to provide them with a short bursts of alertness that will allow them to find a safe place to stop driving and take a break. These wider applications have the potential to reach tens of millions of customers and indeed anyone who could benefit from improved alertness and performance.
Task Progress:	Experiment design - We have used Prof. Elizabeth Klerman's CPSS model to model the protocol and explore the effects of individual differences, e.g., circadian period, to evaluate the effects on cognitive throughput. Drs. Lockley and Brainard specified the light spectrum to be used in the experiments. The experiment design was presented at the NASA Human Research Program Workshop in February 2010. There was considerable interest in the proposed robotics tasks since they have high face validity to space operations and are much higher level tasks than commonly used in fatigue research. We will address the question whether performance on simple vigilance tasks, e.g., PVT, correlates with performance metrics of the robotics tasks. Four staff members have travelled to JSC for Generic Robotics Training. Three MIT staff members have completed the Brigham & Women's Hospital orientation and training to participate the in the in-patient studies. The training was arranged with the help of Paul Williamson and Zakiya Tomlinson. The team has biweekly meetings/telecons. We have had two experiment design reviews (Nov 2009 and Apr 2010). After the second review meeting, we decided to run the Slam-shift protocol first since the effects are expected to be larger under these conditions. Most of these subjects will be completed in Year 2. The Sleep Restriction protocol will begin in Year 3 and data analysis and publishing will occur in Year 4. The study has received approval from the Center for Clinical Investigation at BWH and study orders and computerized event schedule programming have been completed. Hardware and software development - Discussions with Williamson and Zakiya Tomlinson about detailed procedures for space teleoperation to be implemented in the simulation - We have developed preliminary versions of 2 of the 3 planned teleoperation tasks (i.e., grappling and payload assembly, track and capture). The third task (autosequence programming task) is under development. The framework for the simulation is based on software from earlier experiments, but additional capabilities such as object grappling and animated objects are being included. We are beginning the normative testing of these tasks in the lab. An undergraduate student is assisting with these tests. The simulation hardware (computer CPUs, monitors, joysticks, cart) has been assembled and delivered to BWH to begin safety check. We developed a custom made translational hand controller to use in conjunction with a commercial 3-axis joystick used as the rotational hand controller. These hand controllers provide the same functionality as the hand controllers used in flight. - The LED light panels have been delivered to the Sleep Lab at the Brigham and Women's Hospital. Light stands to support the panels during the experiments are currently being designed. The hardware and software development is scheduled to be completed by the beginning of July, which is about 2 months behind the original schedule.

Bibliography Type:	Description: (Last Updated: 12/18/2024)
Abstracts for Journals and Proceedings	Oman CM, Liu AM, Natapoff A, Tomlinson ZA, Pontillo TM. "Influence of spatial abilities and fatigue on space telerobotics operator performance." 81st Aerospace Medical Association Annual Meeting, Phoenix AZ, May 9-13, 2010. Aviation, Space, and Environmental Medicine 2010 Mar;81(3):214. , Mar-2010
Abstracts for Journals and Proceedings	Oman CM, Liu AM, Rueger M, Evans E, Natapoff A, Czeisler CA, Barger LK, Lockley SW. "Validation of assessment tests and countermeasures for detecting and mitigating changes in cognitive function during robotics operations." 2010 NASA Human Research Program Investigators' Workshop, Houston TX, February 3-5, 2010. NASA Human Research Program Investigators' Workshop, Abstract Book, February 2010. , Feb-2010
Awards	Oman C. "Wyle-LR Young Bioastronautics Investigator Award, NASA Human Research Program Investigators' Workshop, February 2009." Feb-2009
Awards	Lockley S. "Harvard-Australia Fellowship, Harvard Club of Australia Foundation, July 2009." Jul-2009
Awards	Lockley S. "Taylor Technical Talent Award, Illuminating Engineering Society of North America (IES), July 2009." Jul-2009